Chapter 1

Artificial Ignorance, Epistemic Paternalism and Epistemic Obligations

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Proponents of Personalised Medicine claim that developments in Machine Learning technologies will allow us to make better predictions about individuals' propensities to develop disease, hence allowing for more effective prevention and treatment options (Mesko, 2017; Frohlich *et. al.* 2018). Consider two (of the very many) reasons to be sceptical of such claims. One is that patients will mis-understand personalised predictions, leading them to make worse medical decisions. Another is that increased knowledge of personalised risk scores will undermine citizens' willingness to contribute to socialised healthcare, as "the low risk", will realise that participation is not, in fact, in their best interests. Both concerns, if valid, imply reasons to control the flow of information from personalised medical tests: the first implies that results should be available *only* when they are interpreted and contextualised by intermediary experts, such as trained genetic counsellors; the second implies reasons against allowing test results to be available *at all*. How should we decide when such controls on the flow of information are permissible?

In this paper, I propose an answer to this question in terms of "epistemic obligations" and contrast this approach with one framed in terms of individuals' epistemic freedoms. The latter perspective, is, I suggest, implicit in current debates around "epistemic paternalism", and is problematic because it obscures the complex web of social-epistemic norms which bind and structure communicative practice. Section 1 illustrates the obligation-first approach to the ethics of information control through a study of the "understanding worry" about Personalised Medicine, and Section 2 explores its implications for work on epistemic paternalism more generally. Section 3 extends my approach through an analysis of the solidarity concern about PM in terms of duties of ignorance. The Conclusion suggests that many worries about epistemic paternalism are, in fact, best understood as worries about the possibility of abuse of power. We do need to ask when risks of abuse are serious enough that we

should reject controls on information; what we do not need to do is to justify the control of information *per se*.

EPISTEMIC OBLIGATIONS

One way of thinking about controls on the flow of information is as violations of free inquiry. A different starting point is as enabling fulfilment of our epistemic obligations. This section develops the second, less familiar approach. By "epistemic obligation", I refer to demands of ethical, rather than epistemic normativity: obligations to seek knowledge, understanding (or other epistemic states), which derive from our broader ethical obligations (rather than, say, obligations to believe only on the basis of the evidence). It seems clear that our ethical obligations can generate epistemic obligations; if my son makes a solemn promise to his teacher that he will learn the capital cities of Europe, then he has a *prima facie* obligation to do so. At a first pass, we can distinguish two major kinds of epistemic obligations.

First, *self-directed epistemic obligations* - i.e., obligations to ensure we achieve certain epistemic states (knowledge, understanding, etc.), grounded on our more general ethical obligations: as a parent, I am obliged to find out which foods are suitable for young children; as a voter, I am obliged to be well-informed about the policies of different parties; and so on. Second, *other-directed epistemic obligations* – i.e. obligations to ensure that others are well-informed about certain topics: as a parent, I am obliged to ensure that my children know how to cross roads safely; a physician is obliged to ensure that her patients are well-informed about the risks of some procedure, and so on. Note that we can have *other-directed* obligations which do not match-up with any correlative *self-directed* obligations; for example, a physician might have an obligation to inform her patients, even if her patients have no obligation to ensure that they are well-informed. However, some *other-directed epistemic obligations* are grounded in reciprocal *self-directed* epistemic obligations: for example, a climate scientist may have an obligation to inform voters about climate change, such that they meet their obligation to be well-informed about political matters. I will use the term *epistemic enablement obligations* for this subset of *other-directed* obligations.

We can frame both self- and other-directed epistemic obligations in terms of *informing:* self-directed obligations are obligations to become *informed* about some subject matter; other-directed obligations are obligations to *inform* others. As Neil Manson and Onora O'Neill (2007) have explored in their discussion of informed

consent procedures, informing is a complex activity, bound by both ethical and epistemic constraints. Their key insight for this paper is that good informing should be distinguished from merely providing information; a physician who gives you ten pages of small-font information about a drug, with the fact that it might cause a fatal heart attack hidden on page 7, has not discharged her obligation to inform you of the risks of taking the drug.

Why not? I suggest that good informing requires careful curation of a communicative encounter, tied to an understanding of a subject's epistemic and practical needs and limitations. To make this metaphor more vivid, imagine a museum worker charged with curating an exhibition. Simply stacking a pile of exhibits in a room may be an (apparently) neutral way of presenting the past, but doing so is likely to leave visitors, at best, bemused, rather than enlightened! Rather, a good curator will organise and contextualise exhibits in ways which help visitors understand their significance and value. In turn, the ways in which she does this will be sensitive to her audience's epistemic background and practical needs: for example, information cards designed for children will differ from cards designed for connoisseurs. I suggest that similar considerations apply to communication more generally. For example, maybe a good physician would focus on telling the patient about the heart attack risk and making this salient, even if doing so involves leaving out other, epistemically well-established, claims about the drug. A one-page information sheet might leave out lots of information about the drug – say, that it was manufactured in Switzerland, or that it is precisely 1cm long – but be far preferable as an aide to informing than the 10-page manual which includes this information.

Consider, then, the case of Personalised Medicine. There are long-standing worries that patients find it hard to interpret and understand predictions about their risk of disease (Edwards and Elwyin, 2001). These worries have only intensified as risk-scoring tools have become ever more "personalised"; for example, it seems that when patients are told that, in virtue of possessing certain genomic variants, they have a risk of disease, they tend to adopt a "genetic determinism", i.e. they interpret this information as implying that their risk is fixed (Kaphingst *et al* 2012). Such an assumption is, however, false: for example, even if you have a genetic propensity to heart disease, you can still reduce one's overall heart disease risk through exercise and diet.

What are the implications of our knowledge of how patients (mis)understand personalised risk scores for the regulation of Personalised Medicine? The general framework sketched above suggests that there may be good reasons to require that test results are *only* available when they can be contextualised or explained by trained professionals. To see why, consider two ways in which personalised tests might relate to epistemic obligations. First, if we think that individuals have (broadly) ethical obligations to maintain their health, then they have derivative epistemic obligations to become informed about health risks, and PM tests are one way of discharging these obligations. Second, we might (in my mind, quite rightly) deny that anyone has an obligation to be healthy, but think that there is a general obligation to ensure that everyone has the capability to be healthy, which, in turn, grounds an *other-directed epistemic obligation* – typically borne by medical professionals – to ensure that they are well-informed about health risks; the provision of PM test results is one way of discharging this obligation. Either way, mere provision of uninterpreted results is unlikely to succeed in genuinely informing patients; rather, it is far more likely that uninterpreted results will merely confuse patients. We can, then, think of regulations demanding that test results are always contextualised and explained as ways of enabling individuals to meet their self-directed epistemic obligations or of ensuring that experts' other-directed duties to inform are properly discharged. Of course, this is a truncated version of a complex argument. I hope, though, that my main point is clear enough: thinking in terms of "epistemic obligations" provides a simple and powerful framework for thinking through how worries about patient understanding might justify controls on the flow of information.

EPISTEMIC PATERNALISM

Here is an alternative way of thinking about the policies outlined in the previous section: they place limits on freedom of inquiry – i.e. they prevent patients from accessing raw test results – because such freedom will not be in patents' best epistemic interests. That is to say, they seem to be instances of "epistemic paternalism" (Goldman, 1991; Ahlstrom-Vij, 2013), structurally analogous to cases where, for example, we limit peoples' ability to access alcohol on the grounds that prohibition is in their prudential best interests. The second kind of paternalism is, however, highly controversial, because it seems to interfere with individuals' autonomy. We might think, by analogy, that something similar is true of the first kind of case: that the starting

point for thinking about limits on freedom of inquiry is that they are *prima facie* wrongful, even when they promote epistemic well-being. This kind of set-up seems assumed in some recent work on the concept of "epistemic paternalism"; for example, consider Emma Bullock's claim that (in cases of epistemic paternalism): "there will always be a non-epistemic reason against interfering, namely, the harm it would cause to the individual's personal sovereignty" (Bullock, 2018, 441). As Bullock notes, although discussions of epistemic paternalism often focus on the narrower concept of "epistemic autonomy", such practices may seem worrying because they interfere with our more general ability to inquire as we see fit. Although not how Bullock frames her claim, we can express this concern as that epistemic paternalism infringes our "right to free inquiry", grounded in our more general sovereignty. This approach has a *prima facie* appeal. However, I suggest that it distorts our understanding of socio-epistemic norms. (The end of the section suggests a possible explanation of its initial plausibility).

Consider, again, the example of the physician who curates a consent form to ensure her patients are truly informed of the risks of taking the drug. In constructing this form, the physician might make various, ethically relevant mistakes: for example, she might under-estimate her patients' ability to understand complex information, or she may try to influence her patients' decision through how she frames the information she provides. Good informing is, undoubtedly, an ethically and epistemologically complex activity. Still, once we think in terms of a physician's obligations to ensure that her patients are informed, it seems odd to think that the very practice of curation *at all* stands in need of some special justification, because it violates hearers' sovereignty. (Or, more precisely, thinking this way seems to render the concern ethically empty, because such "violations" are such a ubiquitous aspect of our socio-epistemic lives). Starting from the alternative perspective of personal sovereignty seems to conceive of information as akin to something which is floating around "out there" in the world, which we then try to control. This is a mistake: rather, the creation and flow of information is nearly always already embedded in social contexts and institutions, governed by complex ethical norms. To think that curating the flow of information is *always* problematic is to ignore that the flow of information is *always* already shaped by ethical norms.

Of course, matters become more complex once we move to cases such as the regulation of personalised medical test results, because it can be harder to identify who

counts as a speaker, and, hence, who bears which epistemic obligations. Still, a similar lesson is true: while we can discuss the ethical issues involved in particular decisions about how information should be curated, the curation of information itself is simply a core function of a reasonable regulatory structure.

With these comments in place, we face two questions: which ways of curating information are better or worse? How does this obligation-first approach to thinking about "epistemic paternalism" relate to everyday paternalism? These are complex questions, which I cannot answer fully here. However, both challenges point us to an important worry about curating the flow of information: that it *can* easily involve an (illegitimate) over-riding of personal sovereignty. To explain: for speakers to discharge their obligations to inform hearers, they will have to make certain sorts of assumptions about their audiences. Typically, these assumptions will include not only assumptions about their audience's epistemic capabilities ("will they understand this?"), but, also, their practical interests ("is this something they want to know or should know"?) For example, a physician might assume that her patient wants to know about a heart disease risk but not that a drug was manufactured in Switzerland. However, there may be systematic differences between what a hearer really does or should want to know, and what a speaker thinks her hearer really does or should want to know. For example, in virtue of her religious convictions, a hearer may think that it is particularly important to know whether an operation carries a risk – however small – of requiring a blood transfusion, whereas a speaker might think that the hearer is irrational or misguided in wanting this information. Hence, we face a general problem: systems which allow speakers to discharge their epistemic obligations to inform must allow leeway – to include or exclude certain information – but this leeway can be abused by over-ruling the subject's personal value judgments; a system which allows the physician to make the ethically proper decision not to report the fact that the drug was manufactured in Switzerland may also allow the physician to make the ethically improper decision not to report the potential need for a blood transfusion. Such abuses constitute a clear example of the kind of substitution of judgment which marks paternalistic interference (Begon 2016). An important desideratum on ethical information curation is that speakers avoid such abuses. It is, then, certainly true that practices of epistemic paternalism *can* instantiate the same wrongs which make standard paternalism wrong; however, to think that *all* instances of epistemic paternalism are problematic in the

same way as standard paternalism is to flatten the complex ethical landscape which structures information flow.

EPISTEMIC SOLIDARITY

We can distinguish between a descriptive sense of "epistemic paternalism" – to describe a set of practices which structure the flow of information with the aim of improving subjects' epistemic state – and an evaluative sense – as a set of practices which *prima facie* stand in need of justification. I have suggested that we should structure our thinking about informational control policies around the concept of epistemic obligations, rather than restrictions on epistemic subjects' free inquiry. If so, the descriptive concept is useful – because it describes a set of strategies for meeting our obligations, hence reminding us of the complexities of good informing (John, 2018) – but the evaluative concept is not – curating the flow of information is not always *prima facie* wrongful. Still, one might wonder whether we really should adopt an obligation-first approach; one application is, after all, not an argument. In this section, I suggest a further advantage of starting from epistemic obligations rather than epistemic rights: that doing so provides a more elegant way of thinking about a core topic in debates over information control, the creation and maintenance of ignorance.

Consider, again, the case of Personalised Medicine. One worry is that personalised predictions are easily mis-understood or mis-interpreted by audiences, such that untrammelled access to raw test results might worsen individual and population health. A second worry about improved prediction techniques in general (and, hence, relevant to Personalised Medicine specifically) is that patients might understand personalised medicine all too well; through gaining personalised medical information they may lose solidaristic motivation (Kavka 1990; Launis, 2003; Nuffield Council on Bioethics, 2010, Chap.3). This is a complex concern, but the core thought is simple. Two claims seem *prima facie* plausible: first, that justice demands a form of socialised healthcare, where healthcare provision is, at least partially, funded through general taxation. Second, most of those who contribute to actual socialised healthcare systems do so, at least partly, on the assumption that such contribution is in their own expected interest, because there is a good chance that they will need to access such a system in the future. In turn, the second thought seems to rely on citizens being behind a "natural veil of ignorance", where they lack detailed knowledge of their own future health. Behind this

"natural veil", it is both rational and natural to make an "equal risk assumption": that we are each at roughly equal risk of needing healthcare in the future. Personalised medicine may undermine this "equal risk" assumption; at least some will realise that, given their unique risk pattern they are likely to be "net losers" from contributing to shared systems, and, as such, what seems like a prudent investment will come to seem more like a burdensome subsidy to others. Allowing access to personalised medical results will undermine the solidaristic attitudes necessary for sustaining institutions which are demanded by justice.

Leading writers in the field are sceptical that personalised medicine *must* undermine solidarity (Praisanck and Buyx, 2017). Furthermore, even if these concerns are valid, they could, in principle, be responded to in many ways. For the sake of argument, however, I will assume that these concerns are valid, and that the only way in which they could be resolved would be through banning access to personalised medicine results. How should we assess such an information-control policy?

Clearly, it would not count as an instance of epistemic paternalism, because it would not promote subjects' epistemic interests. Still, the general evaluative framework associated with the concept of epistemic paternalism provides one way into thinking about this policy. Just as there are non-paternalist reasons why we might limit personal sovereignty – for example, we might ban alcohol sales because of the harms caused to others – so, too, we might think of this policy as imposing a limit on individuals' epistemic sovereignty as a way of preventing "harms" (or, strictly, undermining social justice). In turn, just as we might debate whether restrictions on alcohol sales are a proportionate response to the harms caused by consumption, so, too, we might try to assess whether limits on access to personalised predictions is a proportionate response to the dangers to socialised healthcare.

Note that this framing seems to imply that there is nothing interesting or distinctive about the fact that the second policy involves restrictions on *epistemic* autonomy and affects our *epistemic* well-being. Rather, our problem becomes a familiar problem – perhaps the most familiar problem – within moral and political theory: how to balance concerns about individual freedom against concerns about maximising the good. Bullock (2018) has argued convincingly that discussions around epistemic paternalism face a problem: either they assume a *eudaimonistic* view of the value of epistemic well-being, or they collapse into familiar ethical debates. She argues – convincingly to my mind – that the *eudaimonic* option is problematic. My example seems to add more grist

to her mill: it does not seem that debates over information control are fundamentally different from more familiar debates around paternalism merely because they involve "epistemic" states.

However, Bullock herself seems to think that, in framing these problems, we should start from thinking in terms of agents' sovereignty and, hence, free inquiry. In this regard, she shares the same underlying starting-point for thinking about "epistemic paternalism" which I challenged in Section 2. From this starting-point, debates over information control are framed as problems of how to commensurate between (broadly) deontic and (broadly) consequentialist concerns; between the "right" to access personalised data and the "harms" to the community of such access. Such commensuration is notoriously difficult. Interestingly, this problem is even worse if you think that the *eudaimonic* option is plausible: debates over information control involve balancing not only concerns about consequences and sovereignty, but also the quasi-deontic good of epistemic autonomy and a quasi-consequentialist good of knowledge. Regardless of whether they view knowledge as intrinsically valuable, in framing information control policies *as if* they involve constraints on inquiry, standard debates over epistemic paternalism lead us to problems about how to commensurate freedom and the good. I will now suggest that an advantage of the alternative, "obligation-first" approach is that it avoids this problem.

In the first section, I introduced the concept of a self-directed epistemic obligation: i.e. an obligation to acquire or maintain some epistemic state, derived from a more fundamental ethical obligation. Typically, when we think about our epistemic obligations, we think about obligations to *improve* our epistemic situation: for example, obligations to become more informed about certain topics. However, we can also have obligations to *fail to improve* our epistemic situation. For example, in virtue of my more general obligations to respect my friend's privacy, I have an obligation not to read his secret diary which he has left open on the table. I can have obligations to remain ignorant (Manson, 2012). (Note a complexity here: we might think of some of the empirical data which underlies work in epistemic paternalism as also suggesting certain sorts of obligations of ignorance: obligations not to acquire information which we know will mislead us. Knowing that, in virtue of her own epistemic frailties, she is likely to be misled by reading observational studies, a physician might have an obligation not to read such reports. This is a different kind of obligation of ignorance, however: in the diary case, reading the diary really *will* improve my epistemic situation – in a way in

which reading the observational reports would not – but the point is that I *still* should not do so).

In the first section, I suggested that we can also have "epistemic enablement obligations", i.e. obligations to enable others to meet their epistemic obligations. Just as we can have obligations of ignorance, I suggest that others can have obligations to enable us to maintain our ignorance. For example, a third party, who knows that I am insatiably (but improperly) curious about my friend's private life, finds my friend's diary open on the table. She has, I suggest, an obligation to hide away the diary as a way of enabling me to ensure that I meet my obligation of ignorance. (Of course, she may also have an obligation to hide the diary as a way of protecting my friend's right to privacy, but the two motivations are reinforcing, not exclusive).

With this backdrop in place, we have a framework for thinking about the "solidarity" concern about Personalised Medicine, centred around two questions: first, do citizens have obligations – grounded in their broader duties to support just institutions – not to gain more personalised information? Second, if they do have such duties, does the State have correlative duties to enable them in the fulfilment of these obligations? My tentative answer to both of these questions is "yes": I suggest that we have the first obligation because it is typically recognised that our obligation to support just institutions can place limits on our pursuit of our prudential interests (in this case, acquiring potentially useful health information). I suggest that the second obligation follows as a result of States' duties to cultivate citizens' dispositions to advance justice (Macedo, 1992). Of course, both of these claims could be contested, and there may be other, competing concerns which overrule these obligations. I will not discuss these substantive issues here. What I want to suggest, rather, is that reframing the solidarity concern in terms of our epistemic obligations allows us to avoid the tension we identified in the "epistemic paternalism" framework. On my proposed approach, by contrast, rather than conceptualise information control policies in terms of a clash between rights and consequences, we can, instead, think about them as embedded in wider patterns of socio-epistemic obligation. Of course, weighing and assessing different obligations is not an easy task, but it is easier than balancing freedom against the good.

CONCLUSION

In this paper, I have done two things. First, I have made a preliminary exploration of two topics concerning the regulation of personalised medicine, arguing that a proper response to new forms of Artificial Intelligence may be to maintain states of artificial ignorance. Second, I have suggested that, despite its use as an analytical tool, the concept of "epistemic paternalism" is not a useful guide to the evaluative and normative issues concerning information control policies. We should, instead, think in terms of interlocking sets of epistemic obligations. To stress: this is not to say that all processes of information control are ethically fine, with nothing to worry about. Rather, it is to point to a genuine, and unresolved tension: there are excellent reasons why various agents should have the power to control the flow of information, but such powers can easily be abused. For example, there may be ethical reasons why the State should control the flow of personalised information, maintaining a just healthcare system, but, also, good ethical reasons to worry that these powers might be abused to maintain unjustifiable forms of ignorance. As such, there may well be pressing ethical, political and practical reasons to resist controls on the flow of information. We shouldn't, however, confuse those powerful reasons with reasons to think that controlling the flow of information is itself something that requires justification.

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